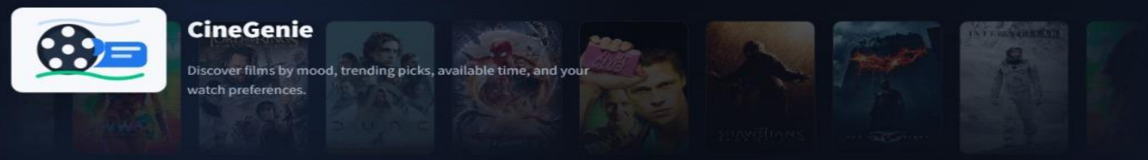
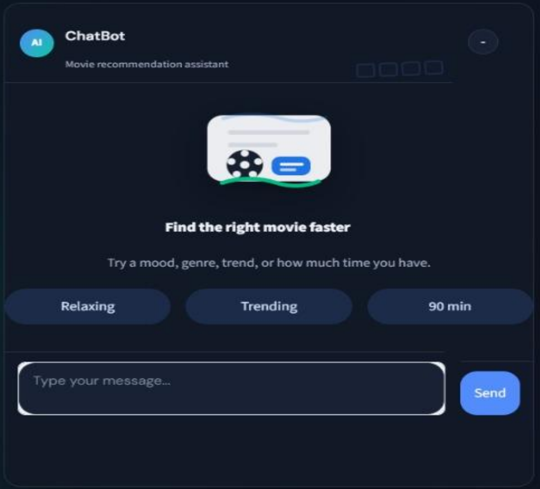


Project # 30
Domain-Oriented Chatbot for Movie Recommendations

CineGenie

Ask . Discover . Watch

Hybrid AI Architecture · Intent Classification · LLM Fallback · TMDB Integration



Contributors:

Source Code: 🎬 https://github.com/pramendraindian/movie-agent

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Architecture Flow

User Input

Streamlit UI / FastAPI /chat

NLP Preprocessing (NLTK tokenize, stem, BoW)

Intent Classifier (NeuralNet · PyTorch)

Recommendation Engine (TF-IDF · Cosine Sim)

TMDB API / LLM Fallback (OpenAI)

Response to User

[For more details, click here ...]

Problem Statement

Users struggle with decision fatigue when choosing from 500,000+ movies, as current tools lack intent understanding, genre/context awareness, and a unified interface for natural language and smart recommendations.

Solution Statement

A unified movie discovery platform that uses natural language with genre- and context-aware dialogue to deliver smart, personalized recommendations and eliminate decision fatigue.

Dataset Description

Attribute	Detail
Source	Kaggle / TMDB API
Dataset	TMDB Movies 2023
Size	930,000+ movies
Format	CSV + API calls
Features	Title, Genre, Rating, Popularity, Overview, Keywords
Preprocessing	Deduplicate, fill nulls, normalize numerics, filter invalid

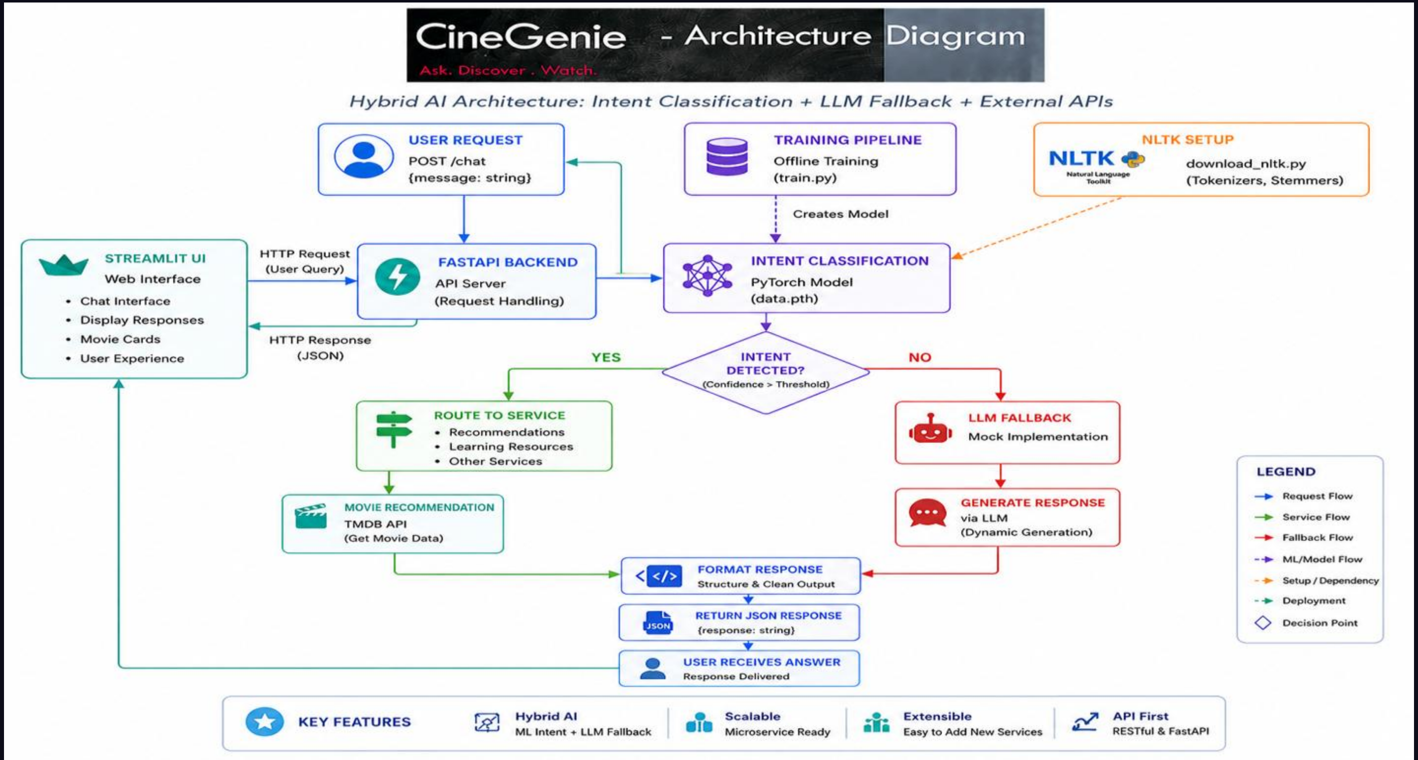
Methodology & System Architecture

Python
Backend

PyTorch
Intent Model

FastAPI
REST API

TMDB
Data Source



https://github.com/pramendraindian/movie-agent

Experimental Details

Neural Network Architecture (Intent Classifier)

Layer	Type	Units / Config
Input	Linear (Dense)	vocab_size → hidden_size (8)
Hidden 1	ReLU Activation	8 neurons
Hidden 2	Linear + ReLU	8 → 8 neurons
Output	Linear (Softmax via CE)	8 → num_tags

Evaluation Metrics

Accuracy ~99%+

Epochs 200

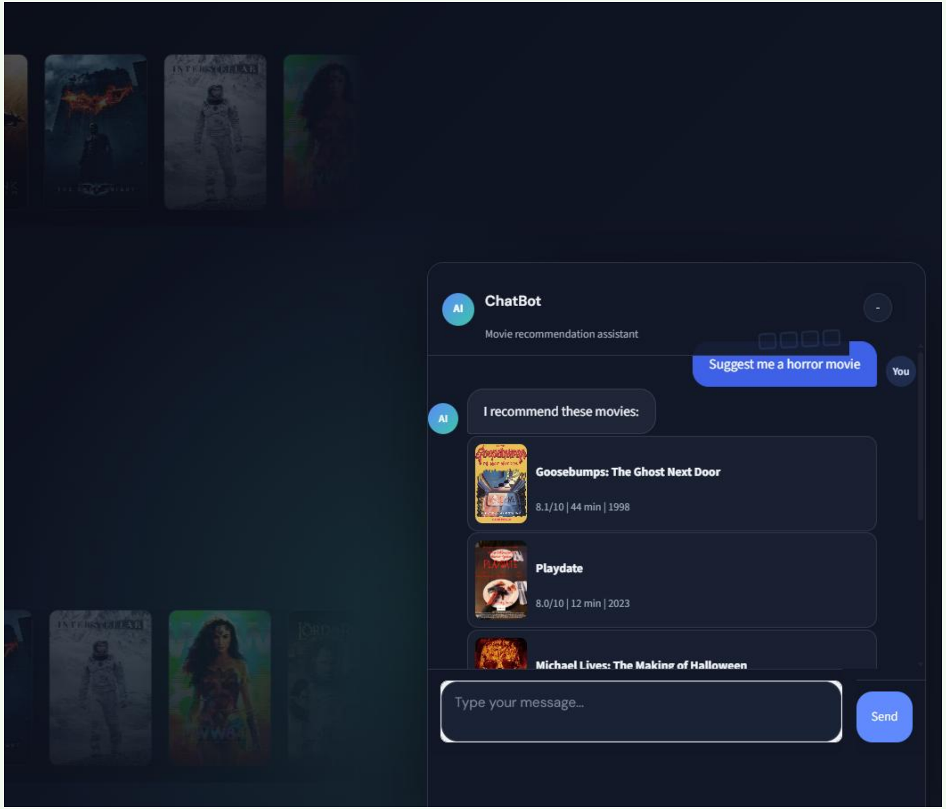
Loss – Decreasing over EPOCHs

API Latency <200ms

```
Epoch [20/200], Loss: 0.7451
Epoch [40/200], Loss: 0.2772
Epoch [60/200], Loss: 0.0668
Epoch [80/200], Loss: 0.0244
Epoch [100/200], Loss: 0.0207
Epoch [120/200], Loss: 0.0197
Epoch [140/200], Loss: 0.0190
Epoch [160/200], Loss: 0.0185
Epoch [180/200], Loss: 0.0180
Epoch [200/200], Loss: 0.0172

Model Accuracy on Training Data: 99.05%
```

Results



Future Improvements

01 **Collaborative Filtering**
Learn from aggregated user ratings and watch history

02 **Hybrid Recommendations**
Blend content-based + collaborative + trending signals

03 **Real-Time TMDB API**
Live movie data instead of static CSV snapshots

04 **User Preference Memory**
Session/persistent state for personalized context

05 **Rating Prediction ML**
Train a model to predict individual user ratings

06 **Multi-Language Support**
Filter & recommend by language and regional content